Clinical Protocol: Comprehensive Type 2 Diabetes & Cardiometabolic Managemer

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Authors:	Division of Internal & Preventive Medicine
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1. Diagnostic Standards & Glycemic Targets

Diagnostic criteria for Type 2 Diabetes Mellitus include:
? Fasting Plasma Glucose (FPG) >= 126 mg/dL (7.0 mmol/L) on two separate occasions.
? Hemoglobin A1c (HbA1c) >= 6.5% using a certified NGSP laboratory assay.
? 2-hour 75g Oral Glucose Tolerance Test (OGTT) >= 200 mg/dL (11.1 mmol/L).
? Random plasma glucose >= 200 mg/dL accompanied by classic hyperglycemic symptoms (polyuria, polydipsia).

Individualized Glycemic Targets:
? Standard adult goal: HbA1c < 7.0% (53 mmol/mol).
? Stringent goal (< 6.5%): Newly diagnosed, young patients without cardiovascular disease, low hypoglycemia risk.
? Relaxed goal (< 8.0%): Elderly patients, extensive comorbidities, history of severe hypoglycemic episodes.

2. Pharmacotherapy: First-Line & Organ-Protective Agents

Modern management prioritizes cardiorenal organ protection independent of baseline HbA1c:

? Metformin:
  - Foundation therapy: Start at 500 mg PO daily with meals, titrate to 1000 mg BID.
  - Renal adjustment: Check eGFR; contraindicated if eGFR < 30 mL/min/1.73m2; reduce maximum dose to 1000 mg daily if eGFR 30-44 mL/min.

? SGLT2 Inhibitors (Empagliflozin 10-25 mg daily, Dapagliflozin 10 mg daily):
  - Strongly indicated in patients with Heart Failure (HFrEF or HFpEF) or Chronic Kidney Disease (CKD) with albuminuria (UACR > 30 mg/g).
  - Reduces cardiovascular death, HF hospitalizations, and renal disease progression.

? GLP-1 Receptor Agonists (Semaglutide 0.5-2.0 mg SC weekly, Dulaglutide 0.75-4.5 mg SC weekly):
  - Strongly indicated in established Atherosclerotic Cardiovascular Disease (ASCVD) or high cardiovascular risk.
  - Substantial weight reduction and significant reduction in Major Adverse Cardiovascular Events (MACE).

3. Hypertension & Lipid Comorbidity Management

? Blood Pressure Target: < 130/80 mmHg.
  - First-line in patients with albuminuria: ACE inhibitor (Lisinopril 10-40 mg daily) or ARB (Losartan 50-100 mg daily).
? Lipid Target:
  - Age 40-75 with diabetes: Moderate-intensity statin (Atorvastatin 20 mg daily).
  - Diabetes + ASCVD or high ASCVD risk: High-intensity statin (Atorvastatin 80 mg daily) with target LDL-C reduction >= 50% and LDL < 55 mg/dL.